

CSE-370 Database Systems

Quiz-2 Set-B

ID, Name and Section :

Time: 20 minutes

Consider a MOVIE database in which data is recorded about the movie industry. The data requirements are summarized as follows:

- Each movie is identified by title and year of release. Each is created by a production company. Each movie has one or more directors and one or more actors appear in it.
- Each movie has zero or more quotable quotes, each of which is spoken by a particular character in the movie.
- Actors are identified by name and date of birth and appear in one or more movies. Each actor has a particular character that they play in the movie.
- Directors are also identified by name and date of birth and direct one or more movies. It is possible for a director to act in a movie (including one that he or she may also direct).
- Production companies are identified by name and each has an address. A production company produces one or more movies.

Draw an EER diagram for the MOVIE database. [10 Marks]

Q2.a. The schema of a Car Repair Shop database is shown below. For each table below identify primary key and foreign keys (if any). For a foreign key, write down which table the key refers to. [5 marks]

Customer

cID	name
1	John Doe
2	Jane Smith

Mechanic

mID	name
1	Alex Carter
2	Jamie Lee

Service

sID	serviceName
1	Oil Change
2	Tire Rotation

VisitToMechanic

cID	mID	time
1	1	10am 12-04-25
2	2	3pm 15-04-25

PrescribedServicesInAVisit

cID	mID	time	sID
1	1	10am 12-04-25	1
1	1	10am 12-04-25	2

Q2.b. Based on your answer to Q2.a, identify the relational integrity constraints that each of the following operations violate (if any). Consider each operation to be independent. [5 marks]

1. Insert <1, 1, 11am 12-04-25> into the Appointment table.
2. Insert <1, 1, 12pm 12-04-25, NULL> into the PrescribedServices table.
3. Delete the second row from the PrescribedServices table.